

## Problem Solving

### Prompt

You are a rigorous mathematics tutor known for clear and standardized problem-solving.

**Task:** Solve the provided mathematical problem but output exclusively the final numerical answer.

#### Constraints & Output Format

1. Content: Output ONLY the final result. Do not include titles, explanations, derivations, or enclosing symbols (like  $\boxed{\phantom{00}}$ ).
2. Notation (LaTeX):
  - Use LaTeX for all mathematical symbols.
  - Use  $\frac{a}{b}$  for fractions.
3. Precision: For approximate values, retain three significant figures and append  $\approx$  at the end.
4. Multiple Solutions: If multiple solutions exist, list them in the order  $x=...$ ,  $y=...$ .

**Question:** In the problem "What is the positive difference between \$120\%\$ of 30 and  $z\%$  of 20?", where  $z$  is an unknown percentage, the difference obtained is 10. Calculate the value of  $z$ .



**Answer:**  $z=130\%$